
nginx-passwd Documentation

Release 4.0.0

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Sep 08, 2026

NGINX-PASSWD DOCUMENTATION

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NGINX-PASSWD

1.1 Overview

Basic Auth Password Manager.

Manages basic auth password files. This replaces the functionality provided by `htpasswd` from Apache. It also provides legacy and modern hash functions, including the `argon2` and `pbkdf2_sha512`.

- All git tags are signed with arch@sapience.com key which is available via WKD or download from <https://www.sapience.com/tech>. Add the key to your package builder gpg keyring. The key is included in the Arch package and the `source=` line with `?signed` at the end can be used to verify the git tag. You can also manually verify the signature

1.2 Recent Changes

Version 4.0.0

- Default hash algo is `argon2`
- Use `meson` / `meson python` for build and package management
- Periodic code review
- tests use (and check depends) on `pyconcurrent` package.

1.3 Note on Hash Functions

We minimize reliance on `passlib` since it is unmaintained and its `bcrypt` functions are broken. Hash functions are separated into *modern*, *active* and *deprecated*. Modern are strongly preferred while active are okay. Deprecated algos should be avoided.

Modern

Algo	Module Used
<code>argon2</code>	<code>python-argon2-cffi</code> (via C library)
<code>pbkdf2_sha512</code>	<code>python-cryptography</code>
<code>pbkdf2_sha256</code>	<code>python-cryptography</code>
<code>bcrypt</code>	<code>python-bcrypt</code>

Active

Algo	Module Used
sha512	passlib
sha256	passlib

Deprecated / Legacy

Algo	Module Used
md5_crypt	passlib
apr_md5_crypt	using passlib

1.4 Getting Started

The nginx-passwd application manages a password file with usernames and their hashed passwords.

1.4.1 Usage

To add or modify a user and write the resulting password file:

```
nginx-passwd -f <password_file> <user>
```

If file is not specified, then the result is written to stdout.

The supported hash algorithms as noted above are:

```
* Modern : argon2, pbkdf2_sha512, pbkdf2_sha256
* Active : sha512, sha256, bcrypt
* Deprecated: md5, apr_md5/apr1
```

with *argon2* the default with version *4.0.0* and newer. In older versions *sha256* was the default.

Older and now deprecated algorithms (*md5*, *apr_md5*) are still supported but really should be avoided and replaced by one of the modern/active ones.

The *apr_md5* algo, also known as *apr1*, is the ancient Apache variant of md5.

If a password is not provided with *-p* option or from stdin, then it will be an empty string.

1.4.2 Options

The options are given below and *nginx-passwd -h* provides a help summary:

```
positional arguments:
  user                  Username

options:
  -h, --help            show this help message and exit
  -f, --passwd_file <file> Password file
  -a, --algo ALGO       Defaults to argon2.
                        * Modern : argon2, pbkdf2-sha512, pbkdf2-sha256
                        * Active : sha512, sha256, bcrypt
                        * Deprecated: md5, apr-md5/apr1
                        Algo names may use "_" or "-"
```

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-p, --passwd PASSWD	Password to use
-D, --delete	Delete this user
-v, --verify	Only Verify password - requires password file

APPENDIX

2.1 Installation

Available on * [Github](#) * [Archlinux AUR](#)

On Arch you can build using the PKGBUILD provided in packaging directory or from the AUR package. To build manually, clone the repo and do:

```
./scripts/do-build  
./scripts/do-install <destination-directory>
```

2.2 Dependencies

- Run Time :
 - python
 - passlib
 - bcrypt
 - argon2-cffi
 - cryptography
- Building Package:
 - git
 - meson
 - meson-python
 - rsync

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